Activity 4.8 – Pipeline Demonstration Monitor

1.Design Approach:

Designed a pipeline demonstration monitoring module using Verilog HDL to observe processor activity across the five pipeline stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM) and Write Back (WB). Implemented a cycle counter to monitor simulation progress.

2.Verification Steps:

- Created a separate testbench.
- Applied reset.
- Provided sample pipeline PC values.
- Checked clock cycle counting.
- Observed pipeline stage signals.
- Verified monitoring operation using waveforms.
- Simulated using ModelSim.

3.Results:

The testbench checks cycle counting and pipeline signal monitoring. PASS/FAIL messages are displayed.

4.Issues Faced:

Pipeline stage signals must be connected correctly. Clock synchronization is important for monitoring. Reset must initialize the cycle counter.

5.Conclusion:

Successfully designed and verified a pipeline demonstration monitoring module using Verilog HDL.